

Heap-Buffer-Overflow Write in Grid Tile Chroma Compositing <= 1.21.2

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Content

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Heap-Buffer-Overflow Write in Grid Tile Chroma Compositing <= 1.21.2

- Advisory: GHSA-frfr-f3vg-2g6j
- CVE: CVE-2026-32740
- Severity: high
- Published: 2026-05-19
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Affected

- `libheif`: <=1.21.2

Description

Summary

A heap-buffer-overflow (write) vulnerability in libheif's grid tile compositing allows an attacker to write **64 bytes of fully attacker-controlled data** past the end of a chroma plane heap allocation by crafting a HEIF/AVIF file with a 1×4 grid of odd-height tiles. The overflow is triggered during normal image decoding with default build configuration. The

written bytes are chroma (Cb/Cr) pixel values from the attacking tile, giving the attacker full control over the overflow content.

Details

Vulnerable function: `HeifPixelFormat::copy_image_to()` in `libheif/pixelimage.cc`, line 1207.

When compositing grid tiles with **YCbCr 4:2:0 chroma subsampling**, the chroma plane offset (`ys`) and copy height (`copy_height`) are computed independently using integer ceiling division (`channel_height(n) = (n+1)/2`). Due to a rounding mismatch, the sum `ys + copy_height` can exceed the chroma plane allocation height by 1 row:

```
// pixelimage.cc lines 1199-1210
uint32_t copy_height = std::min(src_height, channel_height(h - y0, chroma, channel));
uint32_t ys = channel_height(y0, chroma, channel);

for (uint32_t py = 0; py < copy_height; py++) {
    memcpy(out_data + xs + (ys + py) * out_stride,    // 00B WRITE when py = copy_height-1
           tile_data + py * tile_stride,
           copy_width);
}
```

Overflow math (Variant 1, tile_h=65, canvas 64×260):

- Chroma height = 130, allocation = $130 \times 64 = 8320$ bytes
- At 4th tile ($y_0=195$): $ys = (195+1)/2 = 98$, $copy_height = (260-195+1)/2 = 33$
- Last written row = $98 + 33 - 1 = 130$, but max valid row = 129
- **Overflow: 32 bytes per chroma plane × 2 planes = 64 bytes**

The overflow occurs when all of these hold:

1. Grid image with YCbCr 4:2:0 chroma subsampling
2. Odd tile height (e.g., 33, 65, 129, ...)
3. Canvas height divisible by 4 → chroma height is even → no `rounded_size()` padding
4. Canvas height ≥ 128 → chroma height ≥ 64 → no minimum-64 padding
5. ≥ 4 tile rows (rows 1-3 are masked by allocator padding)

PoC

A grid image with:

- 4:2:0 chroma tiles (standard for HEVC/AV1)
- Odd tile height (e.g., 65)
- 4 rows of tiles (so chroma height is EVEN, bypassing `rounded_size` padding)
- $y_0 = 195$ (odd) for the last tile row

ASAN Output:

```
==PID==ERROR: AddressSanitizer: heap-buffer-overflow on address 0xADDR
WRITE of size 32 at 0xADDR thread T56
```

```
0xADDR is located 0 bytes after 8335-byte region [0xBASE,0xBASE+0x208F)
allocated by thread T56 here:
```

```
#0 operator new[]
#1 HeifPixelImage::ImageComponent::alloc() at pixelimage.cc:585
#2 HeifPixelImage::create_clone_image_at_new_size() at pixelimage.cc:2041
#3 ImageItem_Grid::decode_and_paste_tile_image() at grid.cc:532
```

```
SUMMARY: AddressSanitizer: heap-buffer-overflow in memcpy
```

Tester with Two variants with different tile heights (65 and 33) and different allocation sizes (8335 and 4291 bytes) both produce the same 64-byte overflow - proving the attacker can target different heap bucket sizes

Impact

- **Any application** using libheif to decode grid-based HEIF/AVIF files is vulnerable with default build settings
- A crafted `.heic` or `.avif` file triggers writing **64 bytes of attacker-controlled data** past the end of a heap allocation when the image is opened/decoded
- The overflow can corrupt adjacent heap objects (vtable pointers, `std::function`, allocator metadata) → potential **remote code execution** via heap grooming
- **Denial of service** - heap corruption leads to crash on release builds, ASAN abort on sanitized builds

-The attacker can manipulate the tile dimensions to force allocations into specific heap bins (e.g., 4096, 8192). This allows for precise heap grooming to overwrite specific objects like vtable pointers or sensitive metadata in neighboring